



Governing Academic Integrity: Ensuring the Authenticity of Higher Thinking in the Era of Generative Artificial Intelligence

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Abstract

There is a pressing need to reform governance of higher education to ensure the academic integrity in response to the contemporary implications of generative artificial intelligence. The aim of this paper is to articulate research which reveals ways in which this can be done. First, it considers what new information would help academic governors improve their work hence academic practice. It presents an academic integrity indicator framework developed through research reviews, multi-institutional case study analysis, framework and data prototyping, qualitative confirmation with experts, and quantitative analysis. Second, it explores reforms to academic governance to ensure uptake and use of such information. The paper articulates reforms to governance architectures, people and technologies/resources, and the need for external pressure from regulation, benchmarking, and cross-institutional competition. The paper concludes by sketching a research agenda for advancing this field.

Keywords Academic integrity · Academic governance · Assessment design · Governance evaluation

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Digital Disruption of Academic Integrity

Technological transformation is a key driver for universities to remain competitive and relevant in a globalised world, and digitalisation is an important element. Digitalisation involves incorporating digital technologies into core infrastructure and services related to education, research and institution management (Komljenovic et al., 2024). Such digitalisation has long been a source of aspiration and anxiety, particularly for the academic governors who are charged with ensuring the integrity of qualifications. On one hand, digital transformation promotes sophisticated functionality, scalability and skills, greater transparency, and procedural efficiencies. On the other hand, it may prompt dumbed-down work, less in-person engagement, and more breaches of basic academic integrity. How such matters play out has long preoccupied higher education research and development, with the zealotry of marketers often bookended by the cautiousness of academics (Beaton et al., 2022). As this paper articulates (e.g. AdvanceHE, 2024), less energy has been invested in ensuring that academic governance is strengthened in ways which ensure the integrity of academic qualifications.

To this end, this paper brings together research conducted across several countries and years into strengthening academic governance in the context of contemporary digitalisation. It explores the problem of what generative artificial intelligence (GenAI) means for how academic governance must ensure the integrity of qualifications. Academic governance is a remarkably resilient venture, which means that it is both robust and flexible in adapting to new complexities. Yet this paper reveals that contemporary academic governance is not well positioned or poised to handle integrity threats arising from GenAI, in particular those that relate to the authentication of student assessment which sits at the foundation of the integrity of all qualifications. To remedy this situation, the paper advances options for strengthening internal and external governance and poses the challenge of how to make it meaningful across national systems of varying stages of development.

It is important to foreshadow that working on academic integrity and governance is important but sensitive. Handling often murky ethical and legal matters can be tricky, even for experienced academics and managers, and especially when issues are dispersed throughout large and complex institutions (Coates et al., 2022). Such work is particularly fraught given the reputational economy in which universities compete. Indeed, widespread fixation on research performance and rankings, especially when these matters serve as a presumed proxies for education quality, can mislead, distract, disorient or stifle investment in academic integrity. Education excellence and qualification integrity depend on robust, efficient and relevant assessment. There is not yet a flourishing ‘integrity economy’ in higher education. This paper sketches research conducted to open up steps in this direction.

The Case for Governance Reform

It is important to unpack distinctive characteristics of academic governance to frame the empirical and development work which follows. This pinpoints the significance of governing academic integrity, the centrality of student assessment, and the threats and opportunities generated by GenAI.

The ideas and practices of ‘governance’ can mean different things and take many forms in higher education. In its broadest sense, governance is about “the system by which an

entity is controlled and operates, and the mechanisms by which it is held to account” (Governance Institute, 2024). In most higher education systems formal institutional governance is prescribed in a charter or constitution, which gives rise to policies, committees, plans, procedures, forms, and various other instruments and actions (Maassen, 2017). In most universities, in line with widely practised principles of public and private corporation governance, governing boards typically provide oversight to ensure that there are clear lines of accountability for management decisions. Central is the principle that there must be identifiable points of accountability with the board, a clear remit for those responsible for decisions, and no ambiguity regarding authority delegated within the structure of the governance board. In many countries, university governance has evolved in line with new public management (NPM) and marketisation agendas which have given rise to very clear internal hierarchies and identities (Broucker & De Wit, 2015). Internal university governance is accordingly expressed with reference to ‘portfolios’, often with titles like ‘academic’, ‘education’, ‘operations’, ‘finance’, ‘international’, ‘engagement’ and of course ‘research’. This paper focuses on the first two of these (academic and education) stressing, as noted above, that these like the others are invariably blurred, overlapping and dynamic in nature. The rise of generalist managers has blurred many such boundaries and upended traditional dynamics between spheres of authority (Whitchurch, 2017).

While university governance shares much with outside organizations, academic work gives rise to distinctive considerations about who has the authority to participate in governance and the nature of appropriate arrangements. A widespread and often implicit assumption is that by its nature control over generating and disseminating new knowledge must be situated with academic experts. As in any complex field, it is difficult to meaningfully govern ‘good quality education in a discipline’ without the expertise in that discipline sufficient to assess what constitutes ‘good’. This gives rise to distinctive configurations. Due to the nature of academic work, it is often hard to distinguish clearly who the subject and object are because those who are subject to regulations are, themselves, contributors to drawing up and implementing rules. At worst, legally necessary forms of institutional oversight can conflict with the authority that resides in those who participate in the teaching, learning and research that itself defines the university. This challenge is all the more complicated when it incorporates students. As education is a co-produced activity between academic staff and students, seemingly clear divisions of responsibility can falter (see: Gray, 2022). Implicit acknowledgement of this ambiguity often underpins arguments for ‘academic democracy’ (e.g. Trakman, 2008), recognising universities as primarily constituted by communities of staff and students and not merely as places of employment or providers of an education product. The idea that key communities which make up a university should legitimately have oversight of internal operations is common in academic communities. The logic informing this approach is a representational notion that the organisation should be overseen by those with a stake in its operation. A stronger version of democratic participation in higher education often implies that it should be characterised by an enduring, self-sustaining university community, where members share a belief that the principles guiding their association are either shaped by or are the outcome of collective participation.

Notwithstanding complexities infusing academic governance, governing academic integrity remains fundamental. Academic ethics are principles that underpin all academic work and review of policies on Australian university websites conveys that these generally bring into play concepts like honesty, fairness, responsibility, trust and respect. While aca-

ademic ethics are contingent and vary over time and place, many are global in nature and play an inherent role in scholarly communities. Academic integrity means acting in accordance with academic ethics. Academic misconduct, in turn, reflects a contravention of academic integrity. Like ethics, academic misconduct changes over time, across groups, and in different contexts. Higher education is open to an expanding range of academic misconduct threats (Schneier, 2023). These can range from ‘social engineering’, scraping ‘open source intelligence’, ‘phishing’ and ‘impersonation’, to ‘plagiarism’, ‘contract cheating’, ‘exam cheating’, spurious ‘collaboration’, ‘evasion’, ‘impersonation’, ‘AI assistance’ or ‘bribery and coercion’ (TEQSA, 2024a). Universities govern academic integrity and have policies and procedures in place to manage and penalise misconduct. Managing academic misconduct is essential to ensuring academic integrity and assuring the integrity of any higher education qualification.

Academic integrity and misconduct relate to all academic work. Even when limited to education services, integrity covers everything from admissions and orientation, to curriculum and teaching, to student services, and results and graduation. The integrity of assessment is particularly important, however, and is core to the validity of higher education. Universities use assessment to identify, admit, manage and progress students. Irrespective of the characteristics of any specific education program, assessment is invariably the major touchpoint between students and the institution. It lies at the heart and certainly endpoint of any qualification, yielding the information which is aggregated to bestow credentials. Students and whole families live their lives around education assessment and its outcomes. Employers use assessment outcomes to select workers. Governments rely on assessment to develop professions and populations.

Yet the integrity of assessment is under evermore strain. Student assessment has been one of the last frontiers of reform in higher education (Coates et al., 2021). Social spacing arrangements implemented during the pandemic increased the physical and also education and social distance between teachers and learners. Online arrangements put in place spurred higher rates of academic misconduct (Newton & Essex, 2024) and conflation of academic practice (White, 2023). Student approaches to study appear to have become more instrumental, fuelled by the pressures of mass-participation systems, volatile labour markets and national economies, and more vexed migration and geopolitical settings (IOE, 2024). Amid all this change, assessment reform has been vexed, lethargic and devolved, typically and often for understandable expertise-based reasons with little input from institution-level governance.

Contemporary digitalisation appears to have accelerated the sector towards a tipping point. Higher education makes and changes much that shapes the world yet is itself often shaken by forces beyond its control. Generative artificial intelligence (GenAI), like cheap travel and internet, is a case in point. AI has been around for decades, yet GenAI has been quickly and widely disruptive (Lee et al., 2024; Marwala, 2024). Major agencies around the world have scrambled to understand and control it (UK, 2023; OECD, 2023; WB, 2023, Radu, 2021). GenAI offers all kinds of technological augmentation to education when used in smart ways by smart people. Clearly, learning about and using such technology is itself relevant to much academic and professional work. Yet GenAI is only beneficial when used in academically appropriate ways. It is entirely feasible for students to use as yet unregulated GenAI to complete many forms of common assessment, without much individual involvement. Using any such tool as a replacement for thinking can alienate or disengage

students from their academic work, circumventing the kind of thinking or action required for learning.

It follows from the above analysis that there is a pressing need to strengthen the academic governance of the integrity of student assessment in the new era of generative artificial intelligence. The aim of this paper is to chart ways in which this problem can be addressed. The balance of the analysis responds to two research questions. First, what new information would help academic governors improve their work hence academic practice? Universities are inundated with data yet need to ensure they have appropriate data (and use it effectively) to tackle contemporary challenges. Second, what reforms are required to academic governance to ensure uptake and use of such information? The balance of this paper addresses these questions in turn, before the paper concludes by reviewing contributions to research and practical implications.

Developing Academic Integrity Indicators

Though governing minds are free to roam, academic governors ultimately have to rely on matters codified in artefacts such as agendas, papers and reports. Given the challenges to education integrity outlined above, this makes it essential to have available information on academic integrity and related forms of misconduct. What new indicator-level information would help academic governors improve their work hence the integrity of academic practice?

Research and innovation commenced in 2021 to develop governance-level indicator information on the integrity of student assessment. This work evolved out of a larger suite of work on assessment reform and innovation, and predated the late-2022 public release of GenAI (see: Coates & Zlatkin-Troitschanskaia, 2018; Hazelkorn et al., 2018; Zlatkin-Troitschanskaia et al., 2017; Coates & Richardson, 2012; Coates & Seifert, 2010). In synch with much large-scale higher education research, it involved five methodological steps outlined below, including multiyear analysis of research on academic integrity and misconduct, multi-institutional case study analysis, framework and data prototyping, qualitative confirmation with experts, and quantitative analysis with a broader sample of universities across world regions.

Multiyear Review of Existing Research

First, care was taken to ensure an appropriate focus and scope for the research review. The literature search was limited to contributions made over the last decade in globally exposed higher education systems. This scope was chosen specifically to ensure relevance given the complexity and diversity of the topic, and also given that such systems have established themselves as leaders in this area due to high adoption of technology, extensive internationalisation, and the scale. Emphasis was placed on the keywords ‘university’ or ‘higher education’ and ‘generative AI’ and academic governance’. The review was not a passive survey nor an end in itself, but rather an instrument for generating insight into types and dimensions of misconduct, practices including software platforms in common use, legal and educational framings, academic considerations, normative codes and standards, and regulatory activities. Sample works include those by Kier and Ives (2022), Ives et al. (2017),

Dawson (2021), Davis (2023), Cutri et al. (2021), Gallant and Rettinger (2022) and Harrad et al. (2024). Ultimately, much work has been codified in a program of work advanced by the Australian higher education regulator (TEQSA, 2024a). These insights on academic integrity were used to continuously update the existing research.

Multi-institutional Case Study Analysis

Second, case study analysis was conducted over several months during 2022 with five higher education institutions in Australia. These institutions were spread across three Australian states, were organisationally diverse, and were members of a national quality network. This work sought to generate collaboratively affirmed insights into important attributes of assessment validity and integrity. Each institution contributed management personnel, assessment materials, academic experts, students to complete assessment and feedback tasks, and executives to provide formative feedback on reports. This multi-institutional case study work helped to distil a normative structure useful for framing and measuring practice, and specific facets of institutional and educational policy and practice useful for specifying questions to underpin each indicator. It confirmed common institutional practices and cultures, staffing/skill and regulatory pressures, blind-spots and weaknesses, and areas needing support and improvement. Critically, it revealed the lack of information on student assessment and integrity, in particular. This affirmed the governance need and development opportunity.

Framework and Data Prototyping

Third, building on the research and case study analysis, 130 specific items were detailed and organised under eight dimensions of academic integrity. The dimensions themselves were derived from a parameterisation of higher education student assessment. Table 1 shows that these range from considerations pertaining to designing assessment and governance arrangements, developing assessment resources, training and innovation, assessment implementation, the analysis and reporting of assessment data, and evaluation and improvement. The parameterisation is general on purpose so as to contribute governance relevant insights which synthesise educational, institutional, technical and practical considerations across academic disciplines and levels. Table 1 also shows the summary focus of each dimension, the item pool size from which a subset is sampled, and sample items. This version of the framework reflects the culmination of substantial subsequent multifaceted analysis.

Qualitative Analysis with Experts

Fourth, qualitative insight from a range of experts was garnered to ensure that the ideas resonate in practice. To facilitate this, the 130 underpinning items were built into a website data collection and reporting tool. This was distributed via email invitation during 2022 and 2023 to 60 experts, with online feedback received from 56, and in-depth follow-up interviews conducted with nine. These experts spanned European, African, Middle Eastern, Eurasian, Asian, and North/South American higher education systems, represented academics, university vice-chancellors and chief academic officers, evaluation and policy specialists, and experts in academic governance. Table 2 provides a sample of the feedback and derivative changes. Such feedback helped refine the pool of 130 items, calibrate the reporting metric

Table 1 Academic integrity indicator framework and sample items

Dimension	Integrity focus	Item pool	Sample items
Designing	Academic design, governance and management activities	19	Does the institution's top-most board or council receive updates on assessment processes and outcomes? Does the institution have key performance indicators which cover assessment quality and performance?
Developing	Academic and assessment development and production activities	31	What percentage of assessment resembles relevant, real, useful and meaningful problems and situations? What percentage of assessment is designed to have robust academic integrity qualities?
Training	Training and development activities for people involved in assessment	19	What percentage of students are quizzed to determine their knowledge about academic integrity and misconduct? Do induction and orientation practices for new students and teachers include details about academic integrity?
Implementing	Academic and assessment implementation processes and conditions	16	What percentage of assessment uses AI and facial recognition tools used to prevent impersonation and inappropriate activity? What percentage of students are known individually by the teachers who assess them?
Analysing	Data handling, marking and analysis protocols and activities	7	What percentage of assessment is commented using integrated analysis platforms? What percentage of extreme low or high assessment marks are cross-checked?
Reporting	Academic and assessment reporting	10	What percentage of assessment feedback provides learners with specific and actionable advice for improvement? What percentage of assessment is reported in criterion-referenced ways that guide future learning?
Evaluating	Evaluation and detection activities	22	What percentage of assessment is subjected to external audit each year? What percentage of assessment tasks have been reviewed to ensure each contributes unique value and insights?
Improving	Integrity improvement activities	6	What percentage of assessment prompts students to engage in learning? Does the institution have a simple process for referring contract cheating cases?

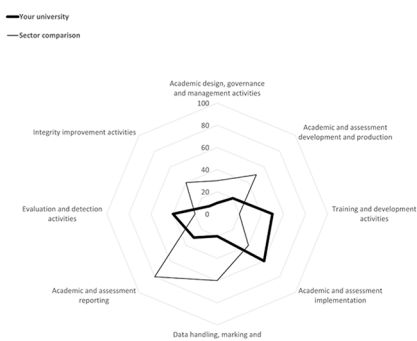
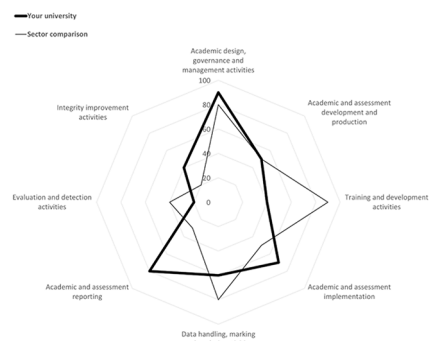
and presentation, and garner broader insights on usefulness and perceived value. Overall, the experts affirmed the need and value of the indicators but queried the potential in a competitive information space hence prospects for dissemination and adoption.

Quantitative Analysis Multiple Universities

Fifth, quantitative data was collected during the expert consultation and from subsequent small-scale application by the institutions which participated in the early case study work and further sample universities. This information was not designed to be representative or generalisable either educationally or statistically, however it did enable testing of the reporting metric. Indicative results are shown in Fig. 1 for just two institutions, each with sector comparisons. Results are scaled onto a 100-point metric, drawing from the framework and items articulated in Table 1. The sector comparisons change given their dynamic calculation due to the use of various contextual covariates including 'institution type', and the low sample size of the pilot. While the depictions are purely quantitative hence difficult to

Table 2 Summary sample expert feedback and resulting changes

Role and region	Summary comments	Sample indicative interpretation
University academic advisor and board member (United Kingdom)	Thank you for sharing the link. It looks really impressive. My main questions (based on our experience at Imperial over the past year) is about how this integrates with other systems. This has been our main challenge in the pivot on-line (and plans going forward). Our assessment systems run through Blackboard, our student information system, Turnitin, often through local assessment processes (including printing or marking on PDFs) and then back into the SIS.	• Clarification of evaluation as a governance instrument • Confirmation of value of the evaluation • Analysis of main enterprise systems for relevance
Academic governance researcher (Europe)	While I do think you ask relevant questions, the problem is that for the respondent, answering them might be a challenge - especially if the response is to be given at institutional level. I do think a series of questions on assessments could be interesting though as a “benchmark” for those in charge of assessments. (Have you done your homework...?).	• Confirmation of multi-level potential of tool • Clarification of use of information as a higher-level indicator
International advisor on higher education (South America)	I would make it clear that adequate assessment is part of smart or effective learning. Too often, we consider curriculum, pedagogy and assessment as three separate dimensions instead of looking at them in an integrated way. How are you planning to use the website? As a commercial venture? Or on a non-profit basis, teaming up with institutions that fight corruption in HE?	• Affirmation of value • Derivative of potential use-cases • Relevance for variety of stakeholders and governance agents
University evaluation specialist and department head (Europe)	...the implicit main message certainly strikes a chord; it's clearly about people and their (information) needs ...layout is not too flashy or designed to cater exclusively for high-end professionals, nice phrasing (not academic!)	• Accessibility of presentation (potentially into languages other than English) • Positioning within broader higher education information contexts
Governance chair and former chief executive (Oceania)	Some overall comments: With the mandating on constructive alignment, including mapping of course learning outcomes, subject learning outcomes and assessment items, you might like to think about some of these questions. I found the coarseness of the responses a little troubling at times. I would want to answer 90% or 80% but could only choose 100% or 75% As per earlier comment, non-self-accrediting providers have their assessment items scrutinised by external expert. Does this count? Similarly, the role of professional bodies.	• Refinement of response scales • Clarification of diagnostic nature of the evaluation • Relevance for different types of providers

Institution 1**Institution 2****Fig. 1** Two illustrative academic integrity indicator institution reports

interpret, they do convey different strategic settings regarding assessment integrity at these two institutions.

Summary Implications

Validated indicator information such as this has the potential to help academic governors improve their work hence academic practice. It would provide governance committees with insight into areas where academics and the institution are well prepared to manage opportunities and threats arising from contemporary forms of academic misconduct. Having data collected across an institution, preferably at the department, discipline or even course level, would furnish insight not only at the institution level but across important contexts and groups. Collecting such data is in itself relatively easy, subject to obstacles often associated with higher education reform. The challenge lies in creating a governance need for such intelligence and having to hand the interpretative intelligence. Notwithstanding the intrinsic interest of governors, pressures arising from contemporary ethical and integrity threats have, as earlier noted, created a tipping point which has an important role to play.

Strengthening Academic Governance

Even if necessary, data is never by itself sufficient to substantiate a position or precipitate reform. Broader institutional and academic substantive framing is required. As conveyed in the above analysis, academic integrity indicator information provides essential information on assessment design and practice. As affirmed by Gupta (2024), institutions as well need to reform academic infrastructure to handle contemporary challenges and opportunities around academic performance and integrity. Institutional governance reforms are required to spur growth of the proposed ‘integrity economy’ in higher education.

Ultimately, what is needed is a means of putting higher education institutions ahead of the people and institutions peddling illegal and unethical education-related services. Interventions are required with governance architectures (policies, committees, etc.), people (students and staff), and technologies/resources (used for administration and learning). This then spurs the need to change specific governance operations. With mind to research on university governance (Shin & Jones, 2022; Hazelkorn et al., 2018) a series of reforms were derived from governance-level work spanning a year including design of a large institution’s governance operations, cross-institutional consultation via benchmarking and collaborative research, and analysis of institutional reforms through Australia-wide consultation.

Governance infrastructure would need updating. The cross-cutting nature of academic integrity and misconduct means that a holistic institutional vision or position is required to make clear the philosophical position. Existing committee structures may be sufficient, or they may need to be augmented by standing or working groups with specific expertise and oversight around academic integrity in the era of GenAI. Expertise on integrity and cyber security in particular is essential and a core skill in any governance capability matrix (ACSC, 2024). Policies need to be updated, as do underpinning procedures and guidance notes which most institutions use to bring policy into practice. Far from being a mere ‘vertical’ or ‘enabler’, effective governance of technology would now appear to be an intrinsic part of higher education. Academics and convenors need to revise curriculum and teaching and in particular assessment to embrace and manage GenAI.

Governance architecture changes are merely performative unless an investment is made to train and educate people. This requires dedicated training and resources for all members engaged in governance roles to ensure they understand responsibilities, remits, and processes. Formal courses on AI governance have started to emerge (e.g. CUGR, 2024; Uhl, 2024, AICD, 2024). The central principle here is that these governance bodies exist to provide oversight through a process that supports democratic participation. Key stakeholders in such change include students, academics/teachers, other staff, leaders, and also parties outside the institution including employers, student family/social groups, and ultimately/ideally those involved in peddling problematic resources. Each of these stakeholders brings their own perspectives and interests to this formation activity. Clearly, teachers and markers are integral to progress in this area. Each will have a perspective on misconduct and GenAI. Working with professional and industry associations to ensure people graduate with required technological and professional skills is essential.

Change to governance operations is required. As earlier parts of this paper foreshadowed, assessment reform is imperative, and the nature of contemporary challenges means that assessment reform is also ongoing. Evaluating overarching course and delivery risk is essential, which will be aided by information from the academic integrity indicators. Multifaceted approaches to cultivating integrity and technology literacy education would be a priority. Fittingly, given the nature of the risk, translating an approach from the field of cybersecurity could help. ‘Red teaming’ (PWC, 2024) involves applying an adversarial lens to an organisation whereby a group of people try to hack or break into a process or system, such as the integrity of assessment hence of qualifications. Such ‘ethical hacking’ or ‘penetration testing’ helps expose vulnerabilities and reveal areas in need of development. Ultimately, by moving beyond entrenched thinking and implementing a form of organisational thinking, such a process seeks to close the gap between prevailing practice and policy and emerging or frontier threats and opportunities.

Certain institutions have already been revealed as etching important progress in this area. Based on analysis of submissions from all of Australia’s registered higher education providers, TEQSA (2024b) highlights academic integrity frameworks developed by the University of New England, Victoria University and Griffith University, working groups and committees at Monash University and the University of Western Australia, staff and student training across a host of institutions, an infrastructure transformation projects at Edith Cowan University and Western Sydney University. The United Kingdom’s Quality Assurance Agency for Higher Education (QAA, 2024) note the cases of Loughborough University and Oxford Brookes University, among others. Clearly, much can be gleaned from such cases as evidence accumulates.

Very little institutional development is likely to pay out within even large universities without the right kinds of external affordance. Universities are full of experts, yet the challenges posed by unethical and illegal practices around academic misconduct and by GenAI are global in nature and overwhelming even to the world’s best resourced organisations and governments. Higher education, as well, is invariably difficult to reform, not least due to the resilient nature of academic governance. The limited transparency around academic misconduct makes change particularly fraught. Benchmarking and collaboration thus have essential roles to play, as do pressure from quality and regulatory agencies. Ultimately, as foreshadowed in earlier analysis of Heyneman (2019), generating a reputational economy

based on academic ethics rather than research reputation would appear to be a useful path forward.

Formative progress is being made to spur governance reform. Transparency can be an important change driver in higher education (Coates, 2017). Concerningly, however, the world lacks a recognised forum for discussing this topic. There are meetings about universities, about governance and about GenAI, but seemingly no sustainable forum for convening and advancing analysis of the important intersection of these topics. Usefully, given this void, a series of regulatory agencies have cooperated to form the Global Academic Integrity Network (GAIN) which conveys itself as “a consortium of education quality and integrity agencies worldwide joining forces to fight the rise of commercial academic cheating services targeting students” (2024). Important national steps have been made in certain countries, notably with the United Kingdom’s Quality Assurance Agency for Higher Education (QAA, 2024), Quality and Qualifications Ireland (2024) and Australia’s Tertiary Education Quality and Standards Agency (TEQSA) (2024a). In Australia, for instance, the higher education regulator asked all institutions to report what they have planned to and are doing to tackle the implications of GenAI on academic and qualification integrity, which has spurred an important national dialogue. In the UK, with guidance from an advisory group, QAA has compiled publications and enhancement projects, produced a charter, and produced additional resources to aid universities. The best intentions and concerted efforts of higher education accountability agencies are likely to have muted impact, however, given the lagging nature of reform, the lack of resonance with academic practice, and the truly global nature of the situation. Nonetheless, creating dialogue and surfacing practice is a necessary step in system hence eventual practice reform.

Next Steps for Research and Reform

In the face of major threats to academic quality and integrity, this paper has articulated research conducted to present the need for new indicator information and approaches to academic governance. The higher education sector has embraced and institutionalised digitalisation and GenAI, perhaps as much if not more than any industry sector. So far, universities have seen-off catastrophic or existential disruption. Such history portends that GenAI will follow the same path, though as an ‘apex technology’ it is surely a non-ignorable force for change.

The paper has argued that academic governance, which lies at the heart of higher education, needs to adapt, in particular as regards the integrity of the assessment of student learning which lies at the heart of ensuring the authenticity of qualifications. The pandemic socially spaced learners and teachers, and GenAI has facilitated forms of intellectual alienation which have seeded further transactional distance between how learners think and their education. This spurs the need for governance reform, to safeguard against risk and plan for future education transformations. Bringing together ongoing research, this paper has advanced two such changes: the development of novel academic integrity indicators, and the reformulation of associated internal and external academic governance structures.

Reforming higher education is difficult, even for the governments. Yet in light of the above contexts, it is essential to find ways to make higher education respond and be equipped to deal with disruption. New information such as rankings has proven key to such reform

in recent decades. Internal reforms are required, though they can move too slowly given the phenomenon in play. As global research reputation risk indices (i.e. rankings) signpost, stimulating a global reputation race is effective. National and international regulatory agencies have an important role to play in this regard.

While tackling large and growing global issues, this paper has only charted formative conceptual, technical and institutional foundations. These are transboundary, formative and pressing conversations which are starting to emerge (e.g. IFI, 2024; Tallberg et al., 2023). Conceptually, there is a need for ongoing review and analysis to clarify emerging ideas and implications. While robust assessment and academic governance are somewhat enduring in nature, the nature of the digital and disruptive threats continues to change. It is imperative that academic governors across world regions and institutional types are equipped and able to respond accordingly. Technically, there is a need for further validation of the indicators, which involves collecting more quantitative data and subjecting this to psychometric validation. The purpose of these endeavours should be to conduct benchmarking and assist academic governors and policy makers in making informed decisions. Finding ways to integrate the platform with existing enterprise information systems will be critical to generating engagement, change, and spurring a pervasive integrity economy in higher education.

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